WEEK 3 – ICT APPLICATION IN DIFFERENT LIBRARY SERVICES

INTRODUCTION

Information and Communication Technology (ICT) has **transformed the way libraries operate**.
ICT is no longer just about computers; it encompasses **networks, software, databases, communication tools, and digital media**.

This week, we focus on **how ICT is applied in different library services**, making libraries more efficient, accessible, and user-centered.

ICT APPLICATION IN ACQUISITION SERVICES

Acquisition is the process of **obtaining library resources**—books, journals, e-resources, and multimedia materials.

Manual vs Automated Acquisition

Manual: Using paper forms, ledgers, or spreadsheets.

Automated ICT-based: Using software to:

* Create purchase orders

* Track orders

* Monitor suppliers

* Maintain budgets

* Generate expenditure reports

Software Tools for Acquisition

1. Integrated Library Management Systems (ILMS) like **KOHA** or **SLAM**
2. Financial management tools for tracking budgets
3. Email and online vendor platforms for electronic ordering

Benefits of ICT in Acquisition

1. Faster ordering and processing
2. Reduced human error
3. Better record-keeping
4. Budget monitoring and reporting
5. Integration with cataloguing systems for immediate availability

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Example: A librarian can order a new journal electronically and automatically link it to the library catalogue when it arrives.

ICT APPLICATION IN CATALOGUING AND CLASSIFICATION

Cataloguing and classification involve **organising library materials** to make them easily retrievable.

Traditional vs ICT-enabled Cataloguing

Traditional: Cards, manual indexing, hand-written catalogues

ICT-enabled: Computerized catalogues, MARC records, automated subject indexing

ICT Tools for Cataloguing

1. **KOHA Cataloguing Module** – supports MARC21 standards
2. **OCLC WorldCat** – provides copy cataloguing and shared bibliographic records
3. **Library of Congress Authorities** – ensures consistent headings
4. **RDA Toolkit (Resource Description and Access)** – modern cataloguing rules

Advantages

1. Reduces duplication of work
2. Enhances access through OPAC
3. Provides quick retrieval
4. Supports networked cataloguing and resource sharing

Example: When a book is catalogued in KOHA, it is immediately searchable by all users online.

ICT APPLICATION IN CIRCULATION SERVICES

Circulation involves **lending, returning, and reserving library resources**.

ICT in Circulation

Automation supports:

1. Borrower registration
2. Book checkout/check-in
3. Reservation and renewal
4. Overdue alerts (email, SMS)
5. Fine management

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Technologies Used

Barcodes and barcode scanners – speed up check-in/check-out

RFID (Radio-Frequency Identification) – supports self-service and theft prevention

ILMS Circulation Module – manages all circulation activities

Benefits

1. Faster service for users
2. Accurate tracking of borrowed items
3. Reduced queues and manual paperwork
4. Integration with OPAC for reservations

Example: A student can check a book's availability on OPAC, reserve it online, and pick it up without queuing.

ICT APPLICATION IN SERIALS AND PERIODICALS MANAGEMENT

Serials (journals, magazines, newsletters) are **time-sensitive resources**, requiring careful management.

ICT Tools for Serials

1. Serial control modules in ILMS
2. Electronic subscriptions via vendor platforms
3. Alerts for new issues
4. Online indexing and abstracting databases

Benefits

1. Efficient tracking of subscriptions
2. Automated claims for missing issues
3. Immediate access to e-journals
4. Improved reporting on serial usage

Example: Using a serial control module, a library can instantly notify users when a new issue of a journal arrives.

ICT APPLICATION IN REFERENCE SERVICES

Reference services involve **helping users locate and use information effectively**.

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Traditional Reference

1. Face-to-face interviews
2. Printed bibliographies
3. Card files

ICT-enabled Reference

1. Online reference databases (e.g., EBSCOhost, JSTOR, ProQuest)
2. Email and chat reference services
3. Virtual reference desks
4. FAQs, guides, and tutorials
5. Subject portals and curated web resources

Advantages

1. Provides 24/7 access to reference materials
2. Supports remote users
3. Enhances research efficiency
4. Reduces staff workload

Example: A librarian can answer a user's query via a live chat, while simultaneously sending relevant e-resources links.

ICT APPLICATION IN ONLINE PUBLIC ACCESS CATALOGUE (OPAC)

OPAC allows **library users to search for resources electronically**.

Features of OPAC

1. Search by author, title, subject, keyword
2. Check item availability
3. Reserve and renew books
4. View borrowing history

Benefits of ICT in OPAC

1. Reduces reliance on physical card catalogues
2. Immediate access to library holdings
3. Supports networked searching across libraries
4. Integrates with circulation, cataloguing, and acquisition

Example: Students searching remotely can reserve books for pickup or download e-resources directly.

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ICT APPLICATION IN DIGITAL LIBRARIES AND INSTITUTIONAL REPOSITORIES

Digital libraries store and provide access to **digital content** such as e-books, journals, theses, and multimedia materials.

ICT Tools

DSpace, Greenstone, EPrints – digital repository platforms

Content management systems – WordPress, Drupal for library portals

Digitisation tools – scanners, OCR software

Advantages

1. Preserves fragile or rare materials
2. Provides global access
3. Supports academic research
4. Facilitates knowledge sharing

Example: Theses from a Nigerian university can be accessed worldwide through an institutional repository.

ICT APPLICATION IN USER EDUCATION AND INFORMATION LITERACY

ICT supports teaching users **how to locate, evaluate, and use information**.

ICT Tools

1. PowerPoint presentations
2. Interactive tutorials and webinars
3. Library websites and portals
4. Online courses and MOOCs

Benefits

1. Reaches more students efficiently
2. Supports self-paced learning
3. Enhances digital literacy
4. Reduces dependency on one-on-one instruction

Example: A librarian can create an online tutorial on "How to Search Academic Databases" and upload it on the library website.

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ICT APPLICATION IN ADMINISTRATIVE SERVICES

Libraries also use ICT for **management and administration**:

1. Staff attendance tracking
2. Financial management
3. Performance reporting
4. Strategic planning and decision-making
5. Generating automated statistics

Example: An ILMS can generate a report showing the most borrowed books in a year, helping management make informed purchasing decisions.

CHALLENGES OF USING ICT IN LIBRARY SERVICES

Even with benefits, ICT adoption faces challenges:

High cost of hardware and software

Frequent power outages in developing countries

Poor network infrastructure

Lack of trained personnel

Resistance to technology by staff

Maintenance and sustainability issues

CONCLUSION

ICT has **revolutionized library services**:

1. Automation of core services
2. Expansion of digital access
3. Support for remote learning and research
4. Efficient management and reporting

Libraries that effectively integrate ICT **meet user expectations, improve efficiency, and remain relevant in the digital age**.

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